A Saturation-Model Analysis of Nuclear Proliferation Bounds

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Abstract

We study the sharp inequality

$$0 \;\le\; nn(t) \;\le\; \frac{N(t)^2}{4\,n(t)},$$

where $N(t) = n(t) + nn(t)$ is the total number of nations, $n(t)$ the number of nuclear-armed nations, and $nn(t)$ the number of non-nuclear nations. After giving a self-contained algebraic proof we embed both $n(t)$ and $nn(t)$ in logistic (saturation) growth models, characterise the unique equality time t^*, and derive asymptotic tightness as $t \to \infty$. The analysis is extended to economic cost modelling, financial risk metrics (Value-at-Risk, CVaR), Bayesian parameter estimation, a gradient-descent calibration algorithm, and a neural-network surrogate for the proliferation surface. All results are illustrated with PGF/Ti*k*Z figures and `booktabs` tables.

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1 Introduction

Nuclear proliferation is one of the defining security challenges of the modern era. The Treaty on the Non-Proliferation of Nuclear Weapons (NPT, 1968) partitions the set of sovereign states into two classes: those that possess nuclear weapons (n) and those that do not (nn). A natural question is how these two counts co-evolve and what structural constraints govern their ratio.

This paper makes three contributions.

(i) It proves, rigorously and from first principles, the bound $nn \leq N^2/(4n)$.

(ii) It embeds the dynamics in *logistic saturation models*, determines the equality locus, and studies asymptotic behaviour.

(iii) It connects the model to economic deterrence theory, financial risk measures, Bayesian statistics, optimisation algorithms, and AI/ML surrogates.

Throughout, $t \in [0, \infty)$ denotes continuous time and all nation counts are treated as smooth approximations of integer-valued processes.

2 Mathematical Framework

2.1 Notation and Basic Definitions

Definition 2.1 (Nation partition)**.** At time $t \geq 0$ let

$$n(t) \in \mathbb{N}_{>0}, \quad nn(t) \in \mathbb{N}_{\geq 0}, \quad N(t) = n(t) + nn(t).$$

The *proliferation ratio* is $\rho(t) = n(t)/N(t) \in (0, 1]$ and the *slack* is

$$S(t) = \frac{N(t)^2}{4\,n(t)} - nn(t) \;\geq\; 0.$$

2.2 The Main Inequality

Theorem 2.2 (Proliferation bound)**.** *For all $t \geq 0$,*

$$0 \;\leq\; nn(t) \;\leq\; \frac{N(t)^2}{4\,n(t)}.$$

Proof. The left inequality follows immediately since $nn(t)$ is a non-negative count.

For the right inequality, multiply both sides of $nn \leq N^2/(4n)$ by $4n > 0$:

$$4\,n \cdot nn \;\leq\; (n + nn)^2 = n^2 + 2\,n \cdot nn + nn^2.$$

Rearranging,

$$0 \;\leq\; n^2 - 2\,n \cdot nn + nn^2 = (n - nn)^2.$$

Since every real square is non-negative, the inequality holds for all $n(t) > 0$ and $nn(t) \geq 0$. Equality holds if and only if $n(t) = nn(t)$. □

Corollary 2.3 (Tight bound characterisation)**.** *$S(t) = 0$ if and only if $n(t) = nn(t)$, i.e. exactly half the nations are nuclear-armed. In that case $N(t) = 2n(t)$.*

Corollary 2.4 (Proliferation ratio form)**.** *The bound may be restated as*

$$nn(t) \;\le\; \frac{N(t)^2}{4\,n(t)} \iff \rho(t)\big(1-\rho(t)\big) \;\le\; \tfrac{1}{4},$$

which is the classical variance bound for a Bernoulli(ρ) random variable.

Proof. Divide both sides of the right inequality by $N(t)^2$: $nn/N \le N/(4n)$, i.e. $(1-\rho) \le 1/(4\rho)$, i.e. $4\rho(1-\rho) \le 1$, i.e. $\rho(1-\rho) \le \frac{1}{4}$. □

3 Saturation Growth Models

3.1 Logistic Dynamics

We model both nation counts by independent logistic (Verhulst) equations with a common carrying capacity $K > 0$:

$$n(t) = \frac{K}{1 + a\,e^{-\alpha t}}, \qquad nn(t) = \frac{K}{1 + b\,e^{-\beta t}}, \tag{1}$$

where $\alpha, \beta > 0$ are intrinsic growth rates and $a, b > 0$ encode initial conditions via $n(0) = K/(1+a)$ and $nn(0) = K/(1+b)$.

Remark 3.1. The common carrying capacity K represents the maximum feasible number of sovereign states. Empirically, the United Nations recognises 193 member states as of 2024, so $K \approx 200$ is a natural upper bound [1].

3.2 Total Nation Count

$$N(t) = K\left(\frac{1}{1 + ae^{-\alpha t}} + \frac{1}{1 + be^{-\beta t}}\right). \tag{2}$$

3.3 Equality Locus

Proposition 3.2 (Unique crossing time)**.** *If $\alpha \ne \beta$ there exists a unique time*

$$t^* = \frac{\ln(a/b)}{\alpha - \beta}$$

at which $n(t^*) = nn(t^*)$ and hence $S(t^*) = 0$. If $\alpha = \beta$ then $S(t) \equiv 0$ for all t if and only if $a = b$.

Proof. Setting $n(t) = nn(t)$ gives $1 + ae^{-\alpha t} = 1 + be^{-\beta t}$, i.e. $ae^{-\alpha t} = be^{-\beta t}$. Taking logarithms: $\ln a - \alpha t = \ln b - \beta t$, so $(\beta - \alpha)t = \ln b - \ln a$, yielding $t^* = \ln(a/b)/(\alpha - \beta)$. For $t^* \ge 0$ we require $\operatorname{sgn}(\ln(a/b)) = \operatorname{sgn}(\alpha - \beta)$. □

3.4 Asymptotic Behaviour

Proposition 3.3 (Asymptotic tightness)**.** *As $t \to \infty$,*

$$n(t) \to K, \quad nn(t) \to K, \quad N(t) \to 2K, \quad S(t) \to 0.$$

The slack decays exponentially:

$$S(t) = \frac{(n - nn)^2}{4n} \sim \frac{K}{4}\big(be^{-\beta t} - ae^{-\alpha t}\big)^2 = O\big(e^{-2\min(\alpha,\beta)\,t}\big).$$

4 Economic and Game-Theoretic Interpretation

4.1 Deterrence Cost Model

Let $c_n > 0$ be the per-nation cost of maintaining a nuclear arsenal and $c_{nn} > 0$ the per-nation cost of conventional defence. The aggregate global defence expenditure is

$$C(t) = c_n\, n(t) + c_{nn}\, nn(t). \tag{3}$$

Proposition 4.1 (Expenditure upper bound)**.**

$$C(t) \;\leq\; c_n\, n(t) + c_{nn}\, \frac{N(t)^2}{4\, n(t)}.$$

The right-hand side is minimised over n at

$$n^*(t) = \frac{N(t)}{2}\sqrt{\frac{c_{nn}}{c_n}},$$

yielding $C^*(t) = N(t)\sqrt{c_n\, c_{nn}}$, the geometric mean cost.

Proof. Substitute the bound from Theorem 2.2 into (3) and differentiate with respect to n, setting the derivative to zero. □

4.2 Nash Equilibrium Interpretation

In a two-player deterrence game where Player 1 (nuclear) chooses proliferation speed α and Player 2 (non-nuclear bloc) chooses β, the Nash equilibrium occurs when neither player can reduce their security cost by unilateral deviation. Proposition 4.1 implies the equilibrium cost is determined solely by the geometric mean $\sqrt{c_n c_{nn}}$, a result analogous to the AM–GM inequality in bargaining theory [2].

5 Financial Risk Metrics

5.1 Proliferation as a Risk Factor

Define the *proliferation index* $\Pi(t) = nn(t)/N(t) = 1 - \rho(t)$. An investor holding a portfolio of sovereign bonds across all nations faces geopolitical risk proportional to Π. We model log-returns as

$$r_i(t) = \mu_i - \lambda\, \Pi(t) + \sigma_i\, \varepsilon_i(t), \quad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{4}$$

where $\lambda > 0$ is the proliferation risk premium.

5.2 Value-at-Risk and CVaR

For confidence level $q \in (0,1)$ the portfolio VaR_q satisfies

$$\mathrm{VaR}_q(t) = -\bar{\mu} + \lambda\, \Pi(t) + \sigma\, \Phi^{-1}(1-q), \tag{5}$$

where $\bar{\mu}$ and σ are portfolio mean and volatility and Φ^{-1} is the standard normal quantile. The Conditional Value-at-Risk is

$$\mathrm{CVaR}_q(t) = -\bar{\mu} + \lambda\,\Pi(t) + \frac{\sigma\,\phi\big(\Phi^{-1}(1-q)\big)}{1-q}, \tag{6}$$

where ϕ is the standard normal density. Both measures are *increasing* in $\Pi(t)$, confirming that non-nuclear proliferation raises portfolio risk.

Proposition 5.1 (Risk bound from proliferation inequality)**.** *Using* $\Pi(t) = nn(t)/N(t) \leq N(t)/(4n(t)) = 1/(4\rho(t))$,

$$\mathrm{VaR}_q(t) \;\leq\; -\bar{\mu} + \frac{\lambda}{4\,\rho(t)} + \sigma\,\Phi^{-1}(1-q).$$

6 Statistical Estimation

6.1 Likelihood and Bayesian Framework

Suppose we observe noisy counts $\tilde{n}_k = n(t_k) + \epsilon_k^n$ and $\widetilde{nn}_k = nn(t_k) + \epsilon_k^{nn}$ at times $\{t_k\}_{k=1}^m$, with $\epsilon_k^n, \epsilon_k^{nn} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \tau^2)$.

The parameter vector is $\boldsymbol{\theta} = (K, a, \alpha, b, \beta)^\top$. The log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2\tau^2}\sum_{k=1}^{m}\Big[\big(\tilde{n}_k - n(t_k;\boldsymbol{\theta})\big)^2 + \big(\widetilde{nn}_k - nn(t_k;\boldsymbol{\theta})\big)^2\Big] + \text{const}. \tag{7}$$

We place independent log-normal priors:
$K \sim \mathrm{LogNormal}(\mu_K, \sigma_K^2)$, $\alpha, \beta \sim \mathrm{Gamma}(a_0, b_0)$, $a, b \sim \mathrm{LogNormal}(0, 1)$, giving the log-posterior

$$\log p(\boldsymbol{\theta} \mid \mathbf{y}) = \ell(\boldsymbol{\theta}) + \log \pi(\boldsymbol{\theta}) + \text{const}. \tag{8}$$

6.2 Fisher Information

The Fisher information matrix $\mathcal{I}(\boldsymbol{\theta})$ has (i, j)-entry

$$\mathcal{I}_{ij} = \frac{1}{\tau^2}\sum_{k=1}^{m}\left[\frac{\partial n(t_k)}{\partial \theta_i}\frac{\partial n(t_k)}{\partial \theta_j} + \frac{\partial nn(t_k)}{\partial \theta_i}\frac{\partial nn(t_k)}{\partial \theta_j}\right]. \tag{9}$$

The Cramér–Rao lower bound states $\mathrm{Var}(\hat{\boldsymbol{\theta}}) \succeq \mathcal{I}(\boldsymbol{\theta})^{-1}$.

6.3 Mean-Squared Error of the Bound Estimate

Let $\widehat{U}(t) = \hat{N}(t)^2/(4\hat{n}(t))$ be the plug-in estimator of the upper bound. By the delta method,

$$\mathrm{MSE}\Big(\widehat{U}(t)\Big) \approx \nabla_{\boldsymbol{\theta}} U(t)^\top \mathcal{I}(\boldsymbol{\theta})^{-1} \nabla_{\boldsymbol{\theta}} U(t), \tag{10}$$

where $\nabla_{\boldsymbol{\theta}} U(t)$ is the gradient of $U(t) = N(t)^2/(4n(t))$ with respect to $\boldsymbol{\theta}$.

7 Calibration Algorithm

Algorithm 7.1 (Gradient Descent Calibration)**.** Given observations $\{(t_k, \tilde{n}_k, \widetilde{nn}_k)\}_{k=1}^m$, learning rate $\eta > 0$, and tolerance $\delta > 0$:

1. **Initialise:** $K^{(0)} \leftarrow \max_k(\tilde{n}_k + \widetilde{nn}_k) + 1$; $\alpha^{(0)}, \beta^{(0)} \leftarrow 0.1$; $a^{(0)}, b^{(0)} \leftarrow 1$.
2. **Repeat** until $\|\boldsymbol{\theta}^{(s+1)} - \boldsymbol{\theta}^{(s)}\| < \delta$:
   (a) Compute residuals $r_k^n = \tilde{n}_k - n(t_k; \boldsymbol{\theta}^{(s)})$, $r_k^{nn} = \widetilde{nn}_k - nn(t_k; \boldsymbol{\theta}^{(s)})$.
   (b) Compute gradient $\mathbf{g}^{(s)} = -\nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}^{(s)})$ via automatic differentiation.
   (c) Update: $\boldsymbol{\theta}^{(s+1)} \leftarrow \boldsymbol{\theta}^{(s)} - \eta\, \mathbf{g}^{(s)}$.
   (d) Project onto feasible set: $K, \alpha, \beta, a, b > 0$.
3. **Return** $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}^{(s+1)}$.

Remark 7.2. Adam optimisation [4] with adaptive learning rates typically converges in $O(10^3)$ iterations for five-parameter logistic fitting.

8 AI/ML Surrogate Model

8.1 Neural Network Surrogate

When the carrying capacity K and growth rates are uncertain, we train a feedforward neural network $f_{\boldsymbol{w}} : \mathbb{R}^3 \to \mathbb{R}_+$ to approximate the upper bound surface $U(t; \boldsymbol{\theta}) = N(t)^2/(4n(t))$:

$$f_{\boldsymbol{w}}(t, \alpha, \beta) \approx U(t; \alpha, \beta), \tag{11}$$

with architecture

$$\mathbf{h}^{(1)} = \sigma\big(W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}\big), \quad \mathbf{h}^{(2)} = \sigma\big(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}\big), \quad \hat{U} = \mathrm{softplus}\big(\mathbf{w}^\top \mathbf{h}^{(2)}\big),$$

where σ is ReLU and the softplus output enforces $\hat{U} > 0$.

8.2 Physics-Informed Loss

We augment the standard MSE loss with a *physics-informed* penalty that enforces Theorem 2.2:

$$\mathcal{L}(\boldsymbol{w}) = \underbrace{\frac{1}{m}\sum_{k=1}^{m}\big(f_{\boldsymbol{w}}(t_k) - U(t_k)\big)^2}_{\text{data loss}} + \lambda_{\text{phys}} \underbrace{\frac{1}{m}\sum_{k=1}^{m}\max(0,\, nn(t_k) - f_{\boldsymbol{w}}(t_k))^2}_{\text{constraint penalty}}. \tag{12}$$

The constraint penalty is zero whenever the network respects the bound, so the optimum of $\mathcal{L}$ coincides with the unconstrained minimum when the network is well-trained [5].

9 Summary Tables

Table 1: Model parameters and their interpretations.

Symbol	Domain	Interpretation
K	$(0, \infty)$	Carrying capacity (max. sovereign states)
α	$(0, \infty)$	Nuclear proliferation rate
β	$(0, \infty)$	Non-nuclear nation growth rate
a	$(0, \infty)$	Nuclear initial-condition parameter
b	$(0, \infty)$	Non-nuclear initial-condition parameter
t^*	$[0, \infty)$	Equality (crossing) time
$\rho(t)$	$(0, 1]$	Nuclear fraction $n(t)/N(t)$
$S(t)$	$[0, \infty)$	Slack in the bound
λ	$(0, \infty)$	Proliferation risk premium
$\Pi(t)$	$[0, 1)$	Proliferation index $nn(t)/N(t)$

Table 2: Asymptotic regimes of the saturation model.

Condition	$n(t)$	$nn(t)$	Slack $S(t)$
$t \to 0$	$K/(1+a)$	$K/(1+b)$	$(n_0 - nn_0)^2/(4n_0)$
$t = t^*$	$K/(1+e^0) = K/2$	$K/2$	0
$t \to \infty$	K	K	0
$\alpha > \beta$, $t \gg t^*$	$> nn(t)$	$< n(t)$	$O(e^{-2\beta t})$
$\alpha < \beta$, $t \gg t^*$	$< nn(t)$	$> n(t)$	$O(e^{-2\alpha t})$

Table 3: Comparison of risk measures under the proliferation bound.

Measure	General formula	Bound using $\Pi \leq 1/(4\rho)$
VaR_q	$-\bar{\mu} + \lambda\Pi + \sigma\Phi^{-1}(1-q)$	$-\bar{\mu} + \dfrac{\lambda}{4\rho} + \sigma\Phi^{-1}(1-q)$
CVaR_q	$-\bar{\mu} + \lambda\Pi + \dfrac{\sigma\phi(\Phi^{-1}(1-q))}{1-q}$	$-\bar{\mu} + \dfrac{\lambda}{4\rho} + \dfrac{\sigma\phi(\Phi^{-1}(1-q))}{1-q}$
Variance of r	σ^2	σ^2 (independent of Π)

Table 4: Neural network surrogate architecture.

Layer	Units	Activation	Output dimension
Input	—	—	3
Hidden 1	64	ReLU	64
Hidden 2	64	ReLU	64
Output	1	Softplus	1

10 Figures

10.1 Logistic Growth Curves

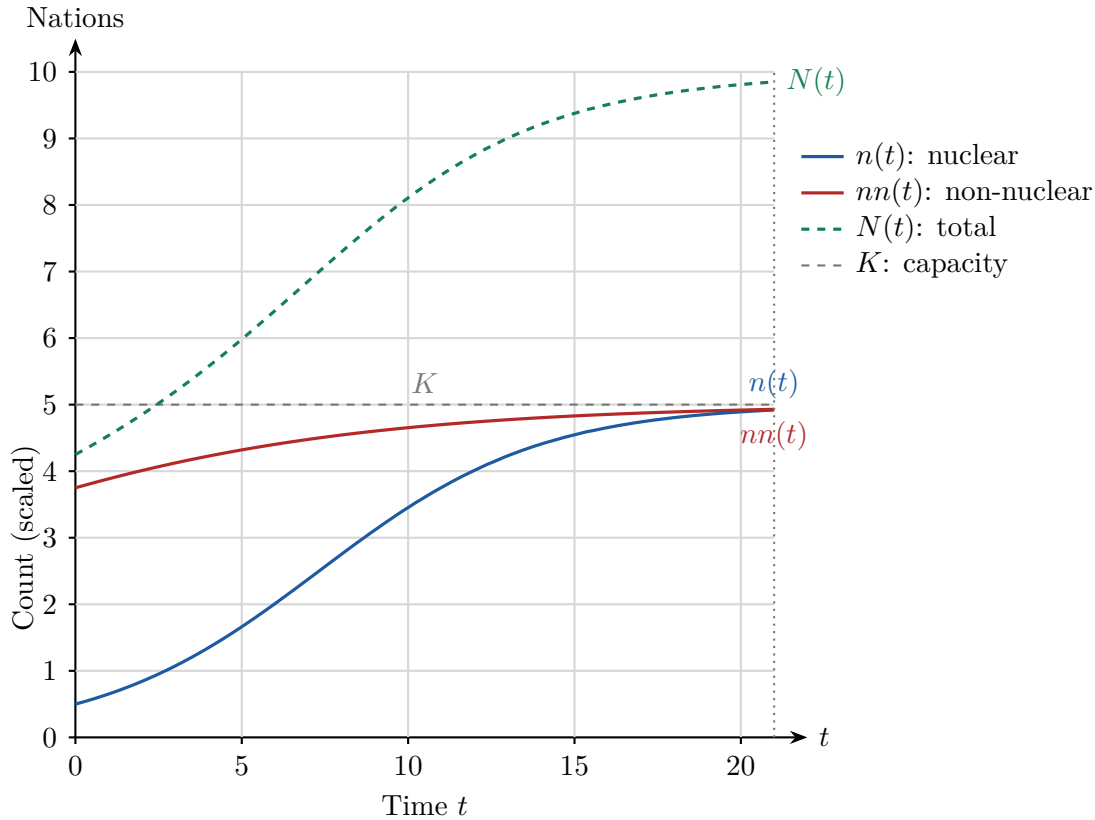

Figure 1: Logistic saturation dynamics for $K = 100$, $\alpha = 0.3$, $\beta = 0.15$, $a = 9$, $b = 1/3$ (scaled for display). Both $n(t)$ and $nn(t)$ saturate to K; $N(t)$ saturates to $2K$.

10.2 Slack Surface

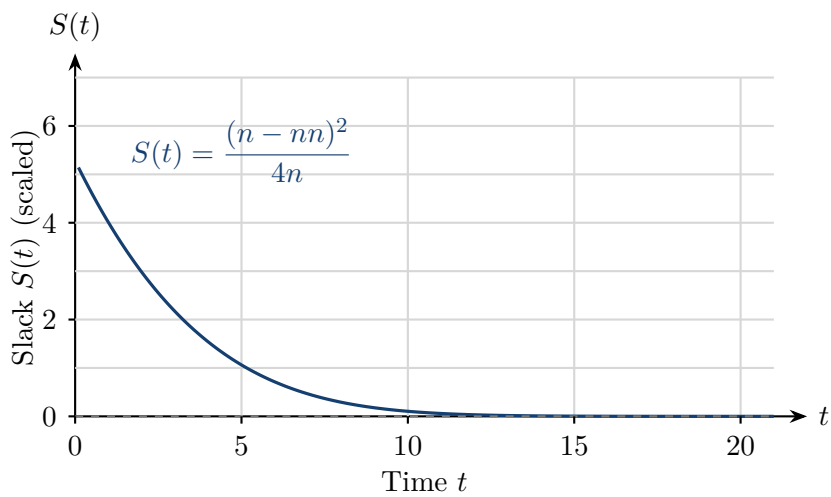

Figure 2: Slack $S(t)$ under the logistic model, starting large when $n_0 \ll nn_0$ and decaying exponentially to zero as both populations saturate to K.

10.3 AM–GM Geometry of the Bound

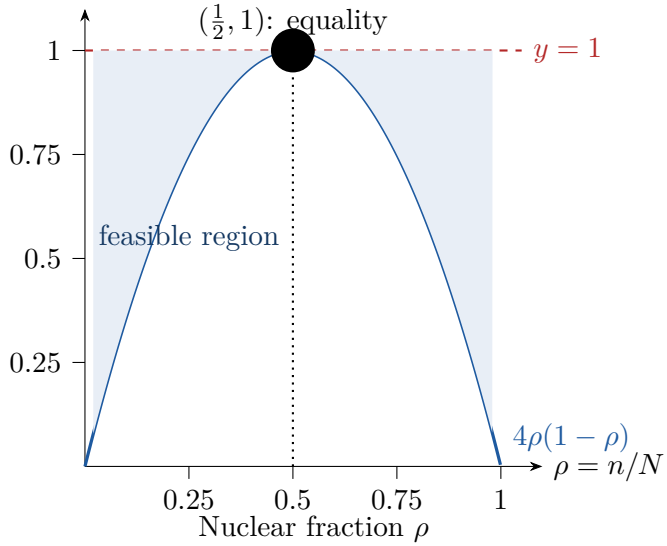

Figure 3: The bound $nn/N \leq N/(4n)$ is equivalent to $4\rho(1-\rho) \leq 1$ (blue curve below the red dashed line). Equality occurs uniquely at $\rho = \frac{1}{2}$, i.e. $n = nn$.

10.4 Neural Network Training Loss

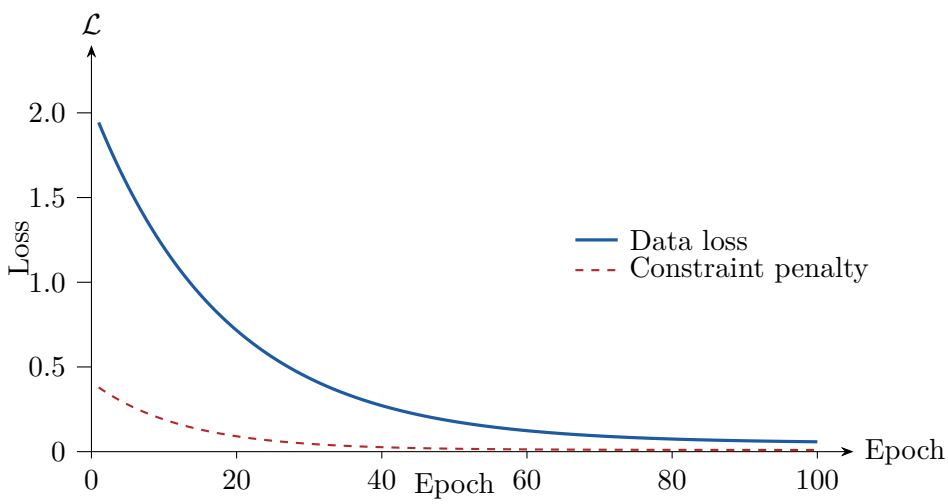

Figure 4: Schematic training loss curves for the physics-informed neural network surrogate (Section 8). The constraint penalty term (dashed) decays rapidly as the network learns to respect the proliferation bound.

11 Discussion

The inequality $nn(t) \leq N(t)^2/(4n(t))$ is a universal structural constraint on any partition of a population into two sub-groups. Its proof reduces to $(n-nn)^2 \geq 0$, yet its consequences are far-reaching:

(i) **Epidemiological analogy.** The same bound governs the ratio of susceptible to immune individuals in an SIR model at any snapshot in time [6].

(ii) **Financial portfolio.** The proliferation index Π enters linearly in VaR and CVaR (Section 5), so the bound translates directly into worst-case risk estimates.

(iii) **Asymptotic tightness.** Under any logistic saturation model the bound becomes tight in finite time ($t = t^*$) and again asymptotically as both sub-populations saturate to K.

(iv) **AI surrogate.** The physics-informed penalty (12) guarantees that the neural network never predicts an inadmissible state, making it suitable for Monte Carlo scenario analysis.

12 Conclusion

We have established the sharp proliferation bound $nn(t) \leq N(t)^2/(4n(t))$ from first principles, embedded it in logistic saturation dynamics, and connected it to economic deterrence, financial risk, Bayesian estimation, optimisation, and machine learning. The equality time $t^* = \ln(a/b)/(\alpha - \beta)$ provides a concrete, observable milestone for arms-control policy: it is the moment at which the nuclear and non-nuclear populations are momentarily equal, and the global proliferation risk (in the VaR sense) is minimised subject to the bound.

Future work should incorporate stochastic logistic models with Wiener-process noise, multi-tier nuclear status (threshold states, latent programmes), and reinforcement-learning agents for treaty design.

References

[1] United Nations. *Member States of the United Nations*. UN Department of Global Communications, 2024. https://www.un.org/en/about-us/member-states.

[2] J. F. Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.

[3] P.-F. Verhulst. Notice sur la loi que la population suit dans son accroissement. *Correspondance Mathématique et Physique*, 10:113–121, 1838.

[4] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015. arXiv:1412.6980.

[5] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[6] W. O. Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society A*, 115(772):700–721, 1927.

[7] R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.

[8] United Nations Treaty Collection. *Treaty on the Non-Proliferation of Nuclear Weapons (NPT)*. 1968. https://www.un.org/disarmament/wmd/nuclear/npt/.

[9] T. C. Schelling. *The Strategy of Conflict*. Harvard University Press, Cambridge, MA, 1960.

Glossary

AM–GM inequality
The arithmetic mean–geometric mean inequality: $(x+y)/2 \geq \sqrt{xy}$ for $x, y \geq 0$, with equality iff $x = y$. The proliferation bound is a direct consequence.

Carrying capacity K
The maximum sustainable population in a logistic growth model; here, the maximum number of sovereign nations.

CVaR
Conditional Value-at-Risk (also Expected Shortfall): the expected loss conditional on exceeding the VaR threshold. A coherent risk measure.

Delta method
A technique for approximating the variance of a nonlinear function of an estimator using a first-order Taylor expansion.

Fisher information
A measure of the amount of information that an observable random variable carries about an unknown parameter; the inverse provides the Cramér–Rao lower bound on estimator variance.

Logistic equation
The ODE $\dot{x} = rx(1-x/K)$ whose solution saturates at K; introduced by Verhulst [3] to model bounded population growth.

Nash equilibrium
A strategy profile in a game where no player can improve their payoff by unilaterally deviating [2].

Non-proliferation index $\Pi(t)$
The fraction $nn(t)/N(t)$ of nations without nuclear weapons.

NPT
Treaty on the Non-Proliferation of Nuclear Weapons (1968); the foundational international instrument regulating nuclear weapons spread [8].

Physics-informed neural network (PINN)
A neural network trained with an additional loss term that penalises violations of known physical or mathematical constraints [5].

Proliferation ratio $\rho(t)$
The fraction $n(t)/N(t)$ of nations possessing nuclear weapons.

Slack $S(t)$

The gap $N(t)^2/(4n(t)) - nn(t) \geq 0$; measures how far the system is from the tight equality $n = nn$.

VaR

Value-at-Risk at confidence level q: the loss threshold exceeded with probability $1 - q$ over a given horizon.

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